Project Development Phase

Data Visualizations

Date	28 February 2026
Team ID	MCA_DA2
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Team Members	Faisal Khan | Divyansh Singhal | Ishita Prajapati | Kali Bakshi

Visualization Summary

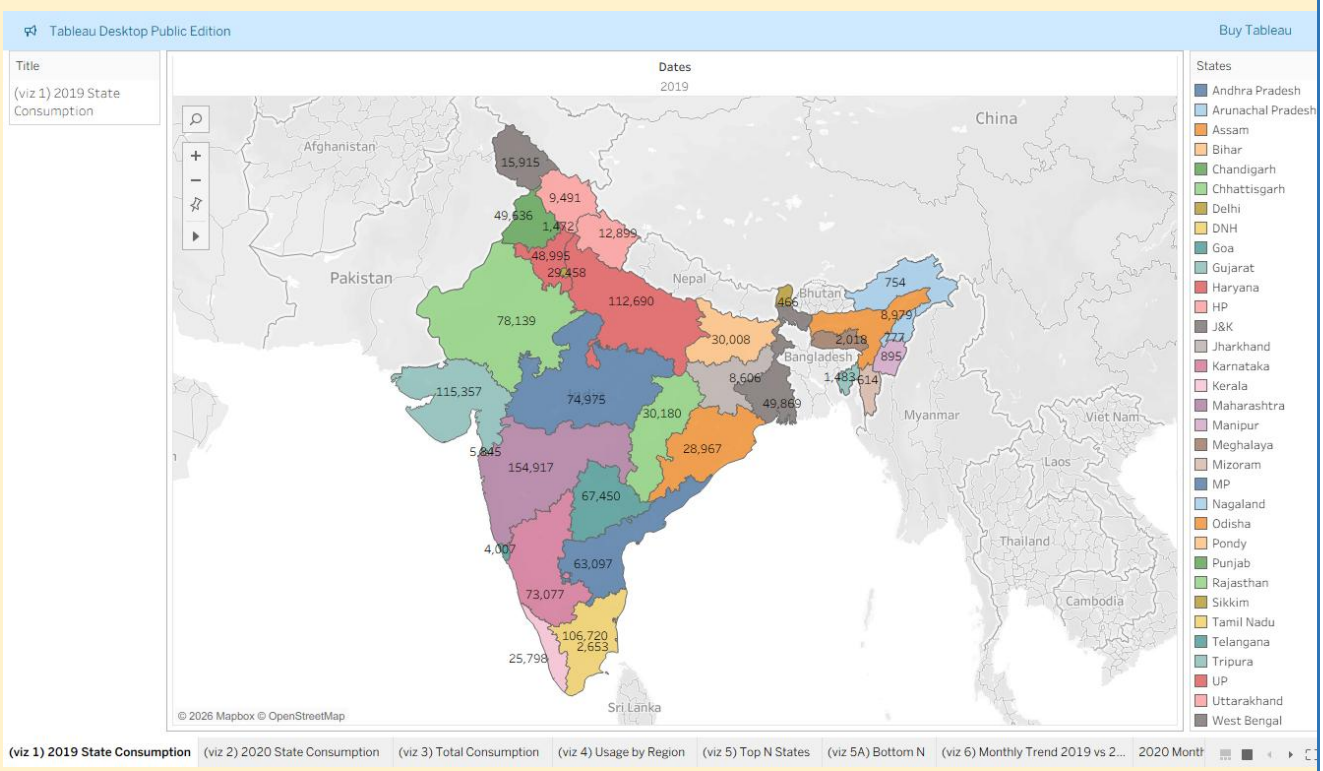
This document lists all 12 unique visualizations created in Tableau for the project. Each visualization is described with its chart type, data used, key insight, and a screenshot placeholder.

Item	Details
Total Unique Visualizations	12
Tool Used	Tableau Desktop 2023 connected to MySQL 8.0
Dataset	Consumption.csv — 16,600 rows, 33 states, 5 regions, Jan 2019 – Dec 2020
Calculated Fields Used	Month Name, Year, Quarter, Lockdown Phase,

Visualization Set 1 — State & Region Overview

Visualization 1 — 2019 State Consumption

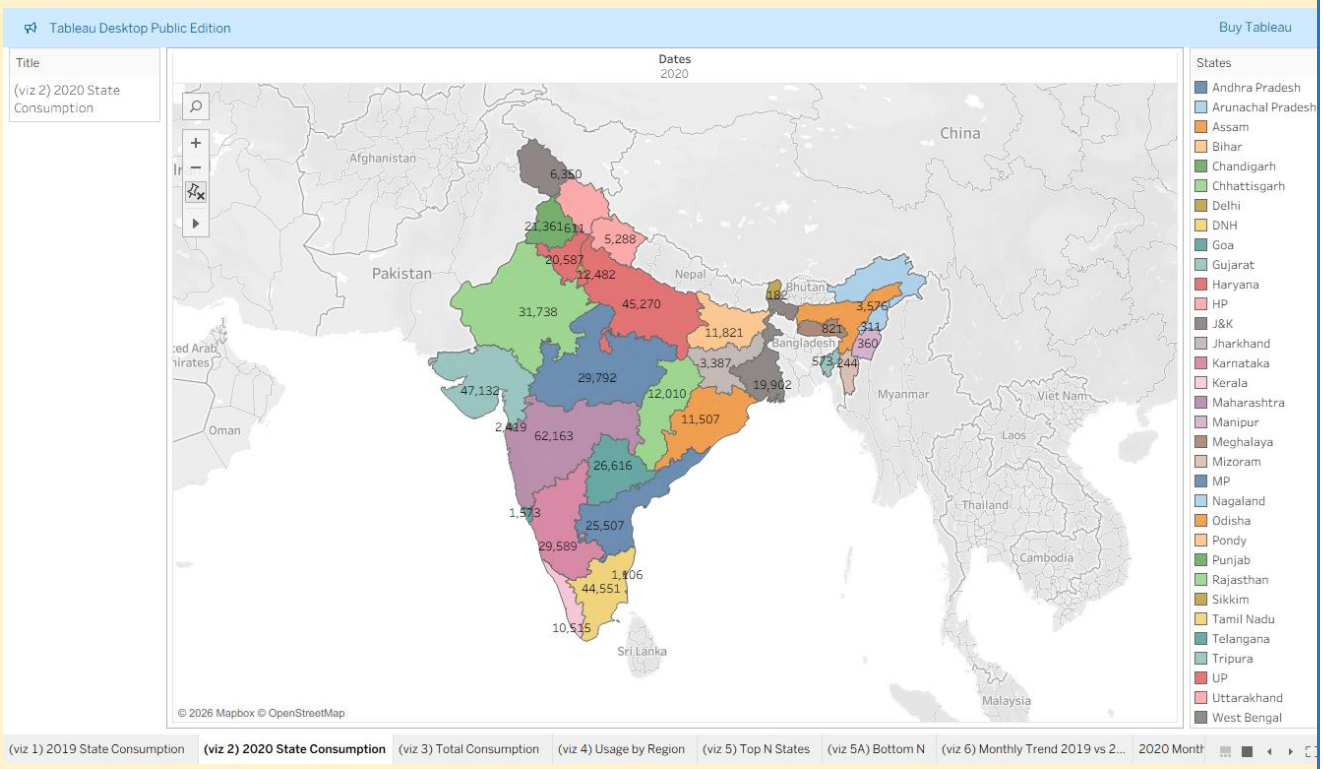
Field	Details
Chart Type	Map Chart
Filter	Year = 2019
Key Insight	Maharashtra leads at 154917 MU. Mizoram is the lowest at 614 MU.
Assigned To	Faisal Khan



Viz 1 — 2019 State Consumption

Visualization 2 — 2020 State Consumption

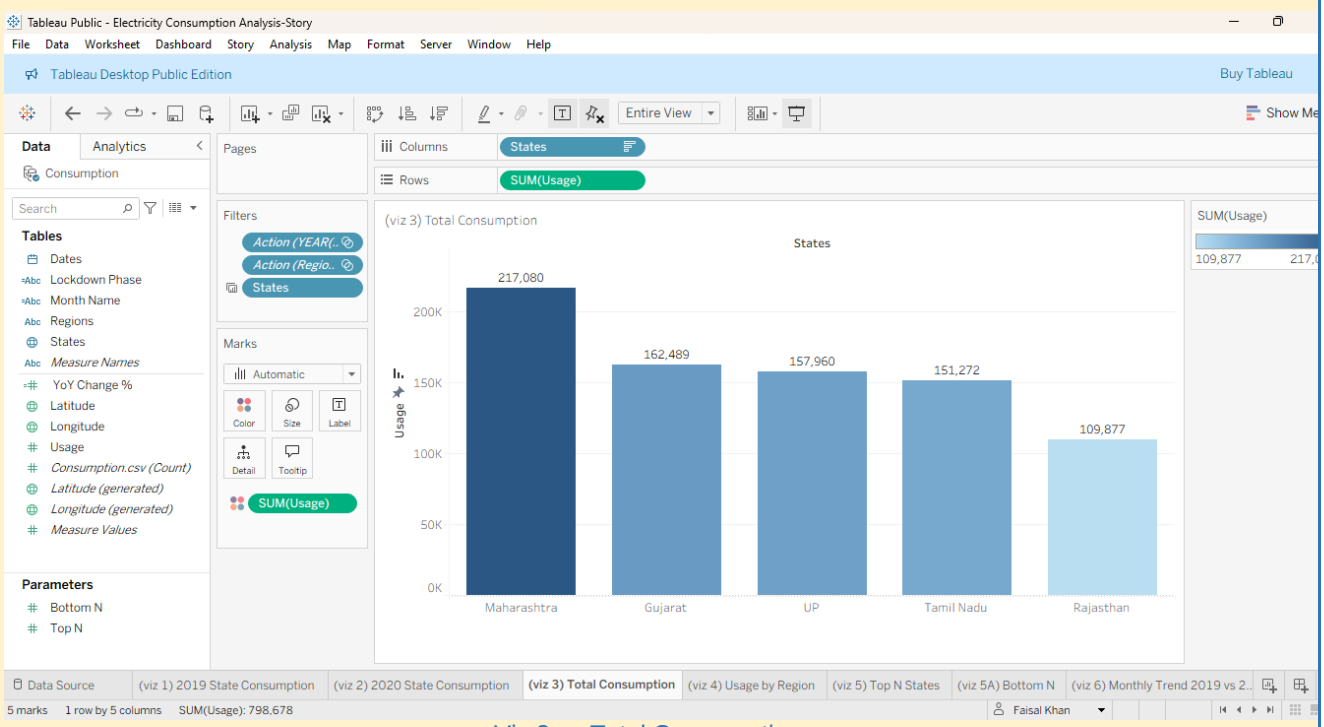
Field	Details
Chart Type	Map Chart
Filter	Year = 2020
Key Insight	All states show lower values vs 2019. Maharashtra still leads at 62163 MU.
Assigned To	Faisal Khan



Viz 2 — 2020 State Consumption

Visualization 3 — Total Consumption (2019 + 2020)

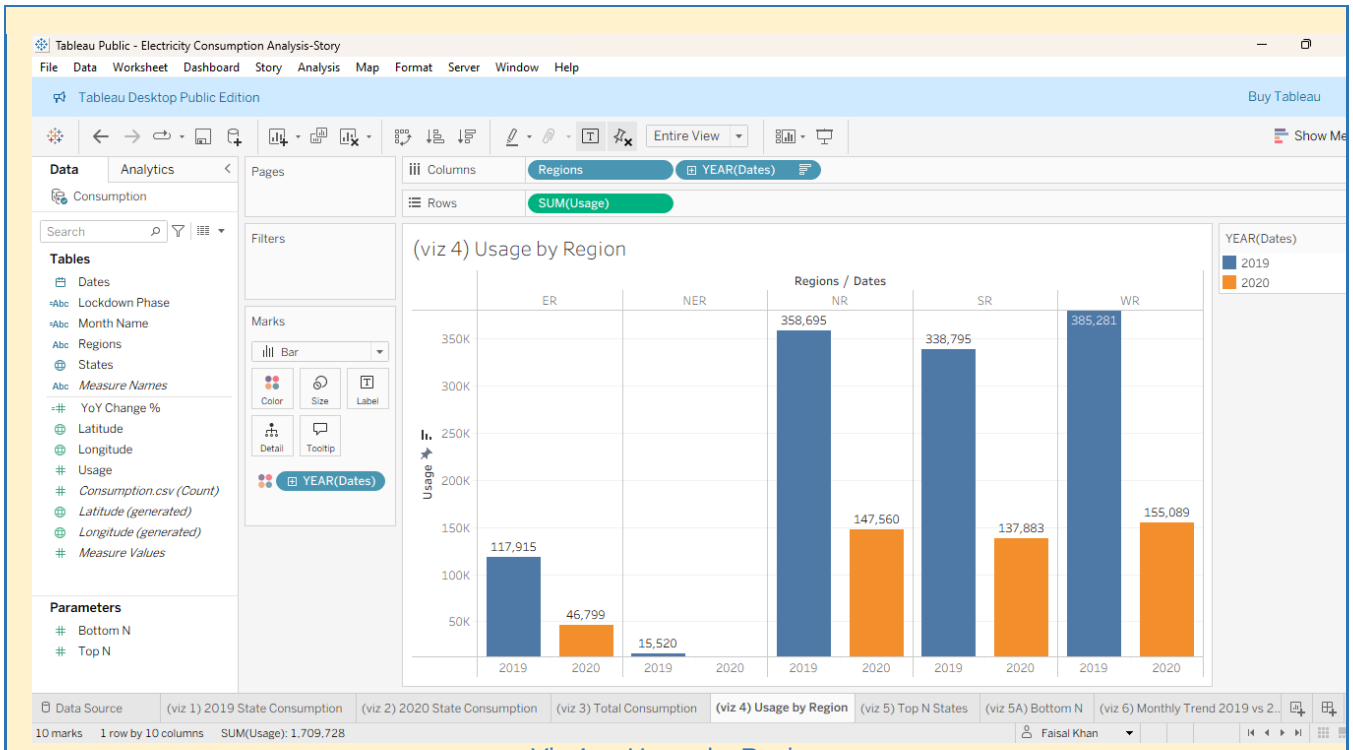
Field	Details
Chart Type	Bar Chart (ranked)
X-axis	States
Y-axis	Total Usage (MU) — both years combined
Key Insight	Top 3: Maharashtra (217,080), Gujarat (162,489), UP (157,960).
Assigned To	Ishita Prajapati



Viz 3 — Total Consumption

Visualization 4 — Usage by Region

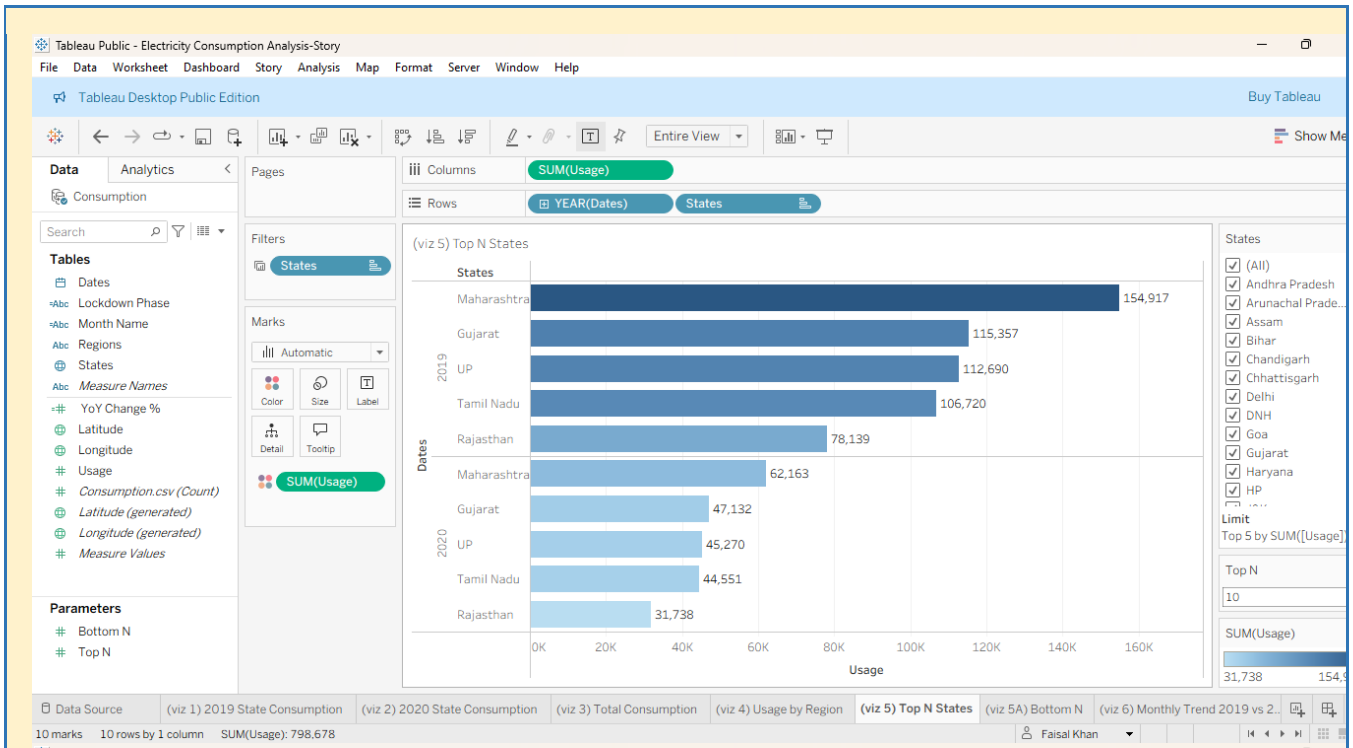
Field	Details
Chart Type	Side – by - Side Bar Chart
Column	Region (NR, WR, SR, ER, NER), Year
Row	SUM(Usage)
Key Insight	WR = 31.2% | SR = 29.3% | NR = 28% | ER = 9.9% | NER = 1.2%
Assigned To	Kali Bakshi

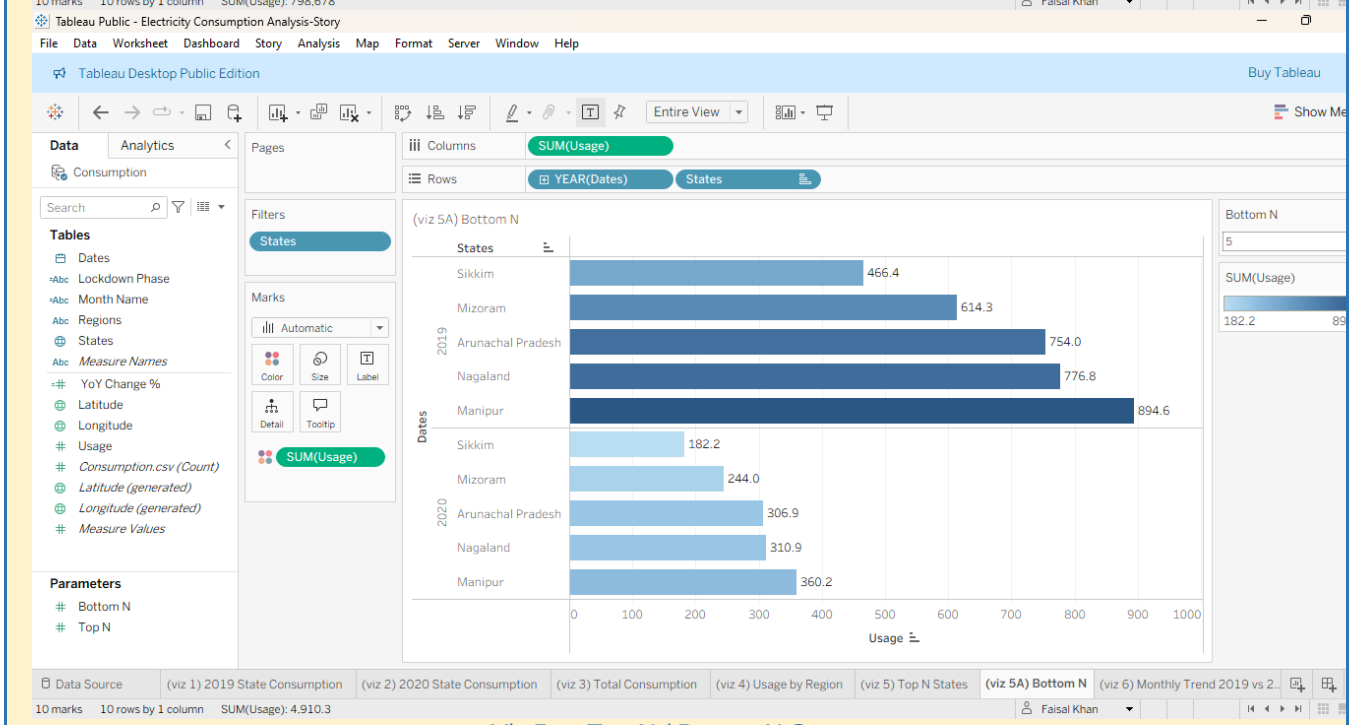


Viz 4 — Usage by Region

Visualization 5 — Top N / Bottom N States

Field	Details
Chart Type	Dynamic Bar Chart with Parameter
Parameter	Top N (1 to 33 states)
Key Insight	Top 5: Maharashtra, Gujarat, UP, Tamil Nadu, Rajasthan. Bottom 5: Mizoram, Arunachal, Nagaland, Manipur, Sikkim.
Assigned To	Faisal Khan



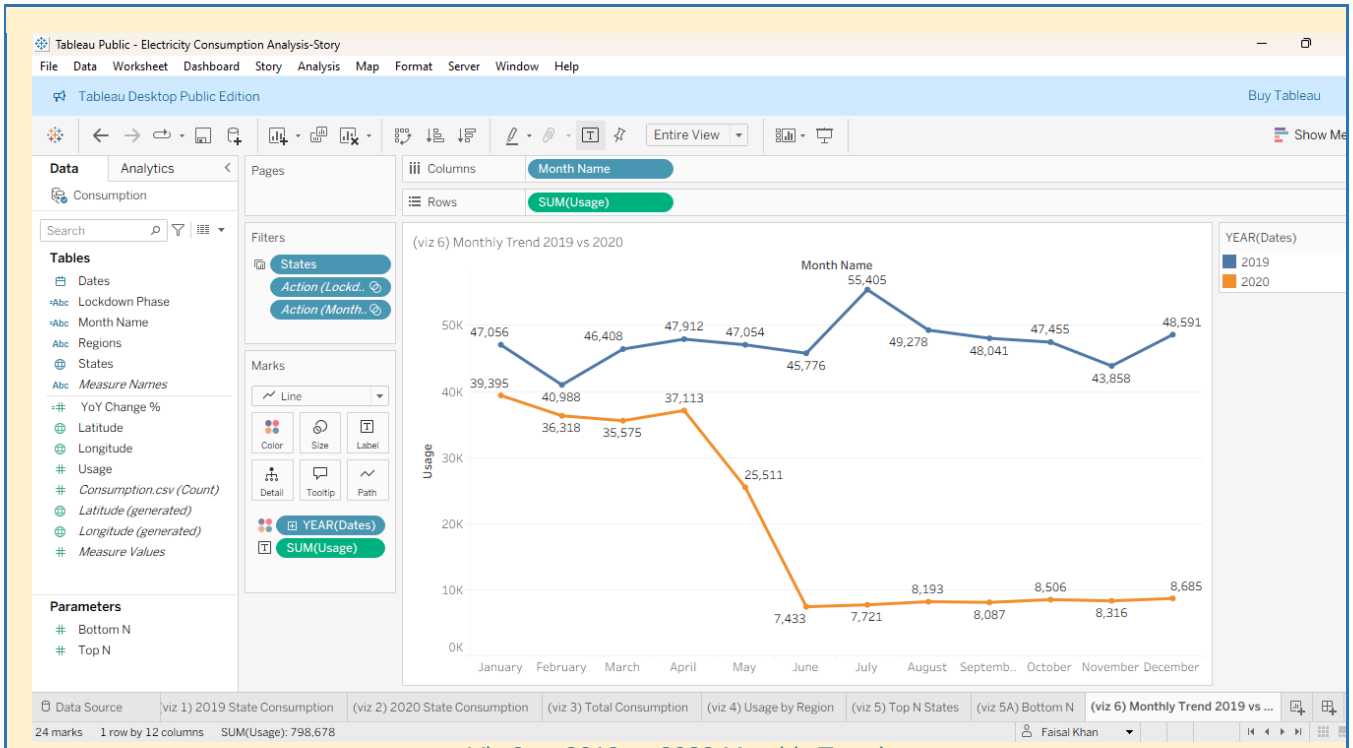


Viz 5 — Top N / Bottom N States

Visualization Set 2 — COVID-19 Lockdown Analysis

Visualization 6 — 2019 & 2020 Month-wise Consumption

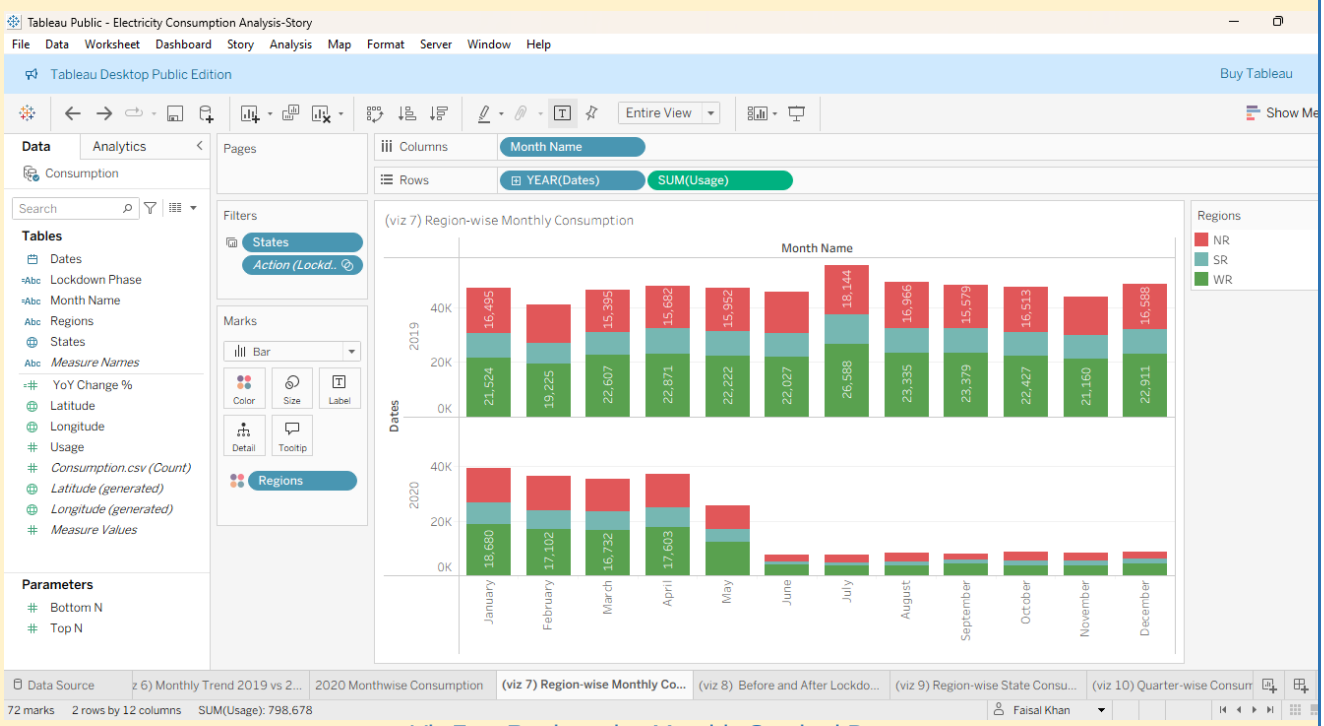
Field	Details
Chart Type	Dual-axis Line Chart
X-axis	Month (Jan–Dec)
Y-axis	Total Usage (MU) — two lines: 2019 (blue) and 2020 (orange)
Key Insight	2020 line drops sharply from March. June 2020 (15,661 MU) is the lowest point — 84% below June 2019.
Assigned To	Faisal Khan



Viz 6 — 2019 vs 2020 Monthly Trend

Visualization 7 — Total Consumption by Region (Monthly)

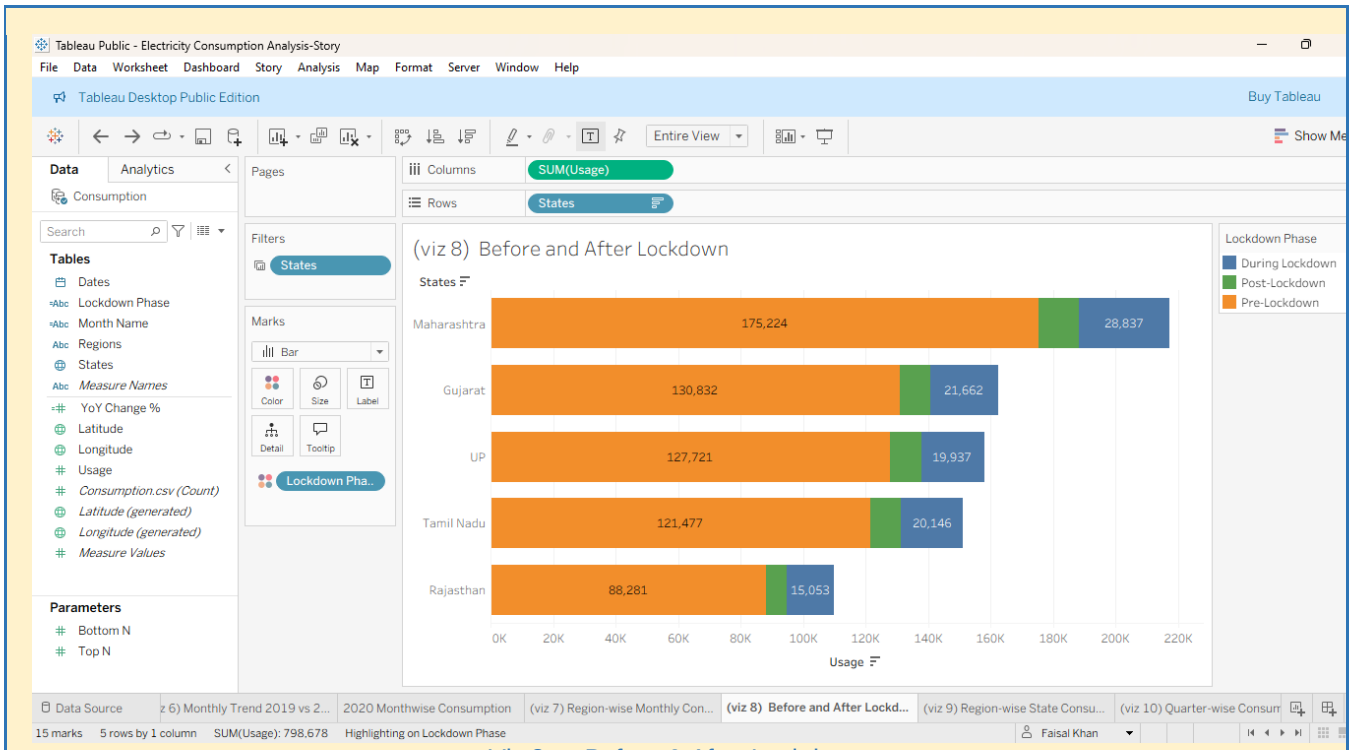
Field	Details
Chart Type	Stacked Bar Chart
X-axis	Month-Year (Jan 2019 – Dec 2020)
Y-axis	Total Usage (MU)
Colour	Region (NR, WR, SR, ER, NER)
Key Insight	WR (orange) is consistently the tallest segment. All segments collapse visibly from March 2020.
Assigned To	Faisal Khan



Viz 7 — Region-wise Monthly Stacked Bar

Visualization 8 — Usage Before and After Lockdown

Field	Details
Chart Type	Grouped Bar Chart
X-axis	Top 5 States
Y-axis	Total Usage (MU)
Groups	Pre-Lockdown (Jan–Feb 2020) vs Post-Lockdown (Jul–Dec 2020)
Assigned To	Faisal Khan

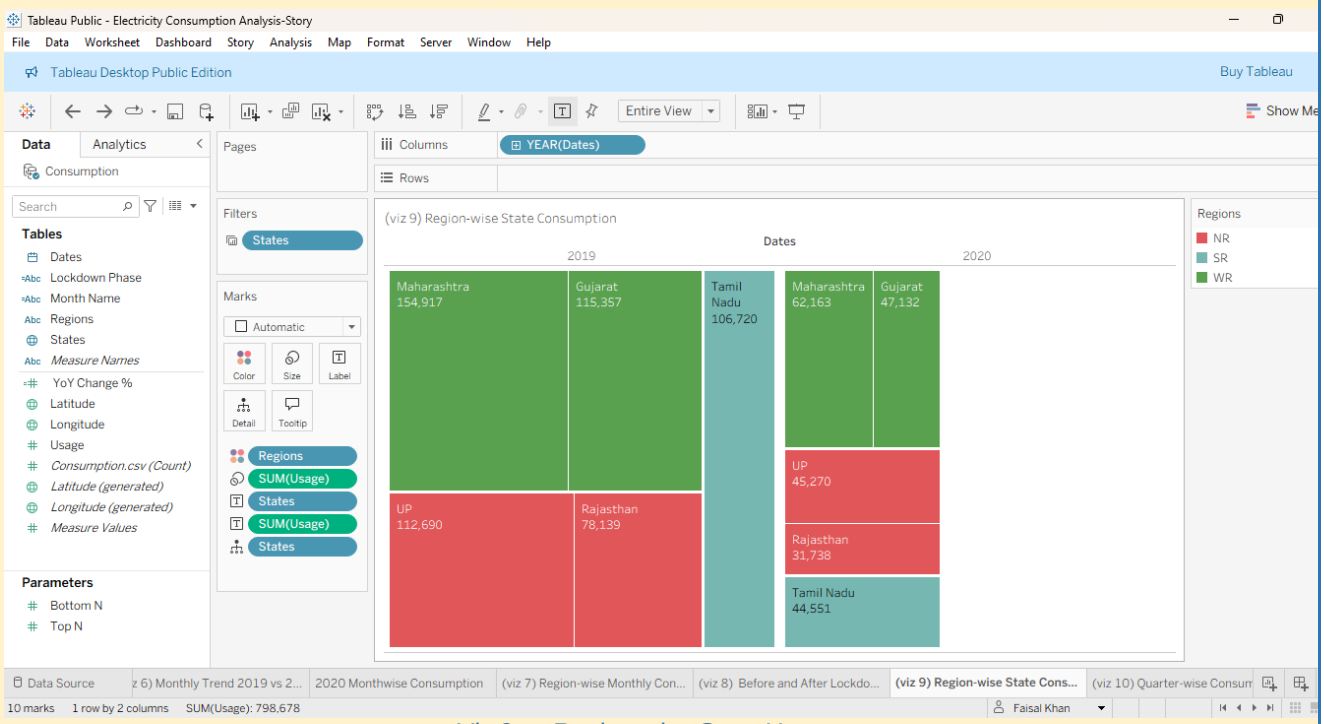


Viz 8 — Before & After Lockdown

Visualization Set 3 — Regional & Time Analysis

Visualization 9 — Region-wise State Usage

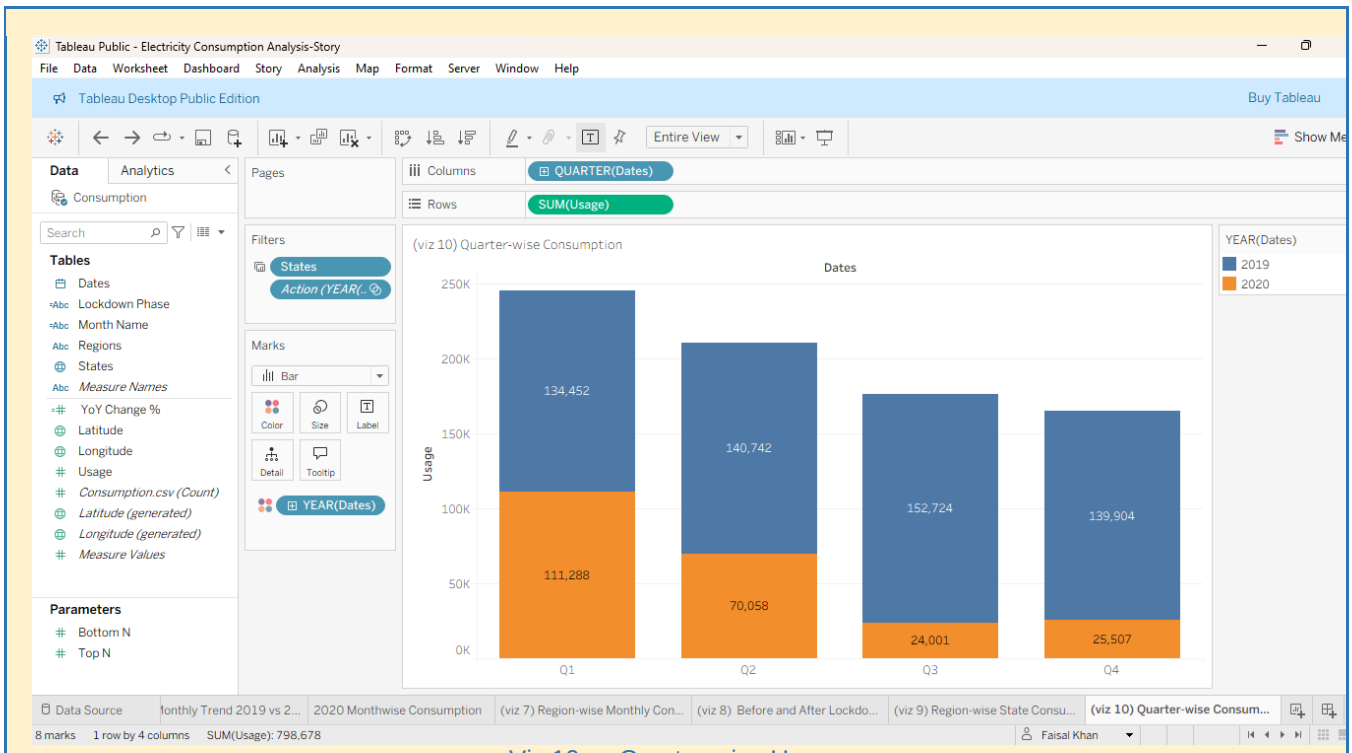
Field	Details
Chart Type	Treemap
Dimension	Region → State
Measure	SUM(Usage)
Key Insight	Within WR, Maharashtra dominates. Within NR, UP is the largest. NER states are all very small.
Assigned To	Ishita Prajapati



Viz 9 — Region-wise State Usage

Visualization 10 — Quarter-wise Usage

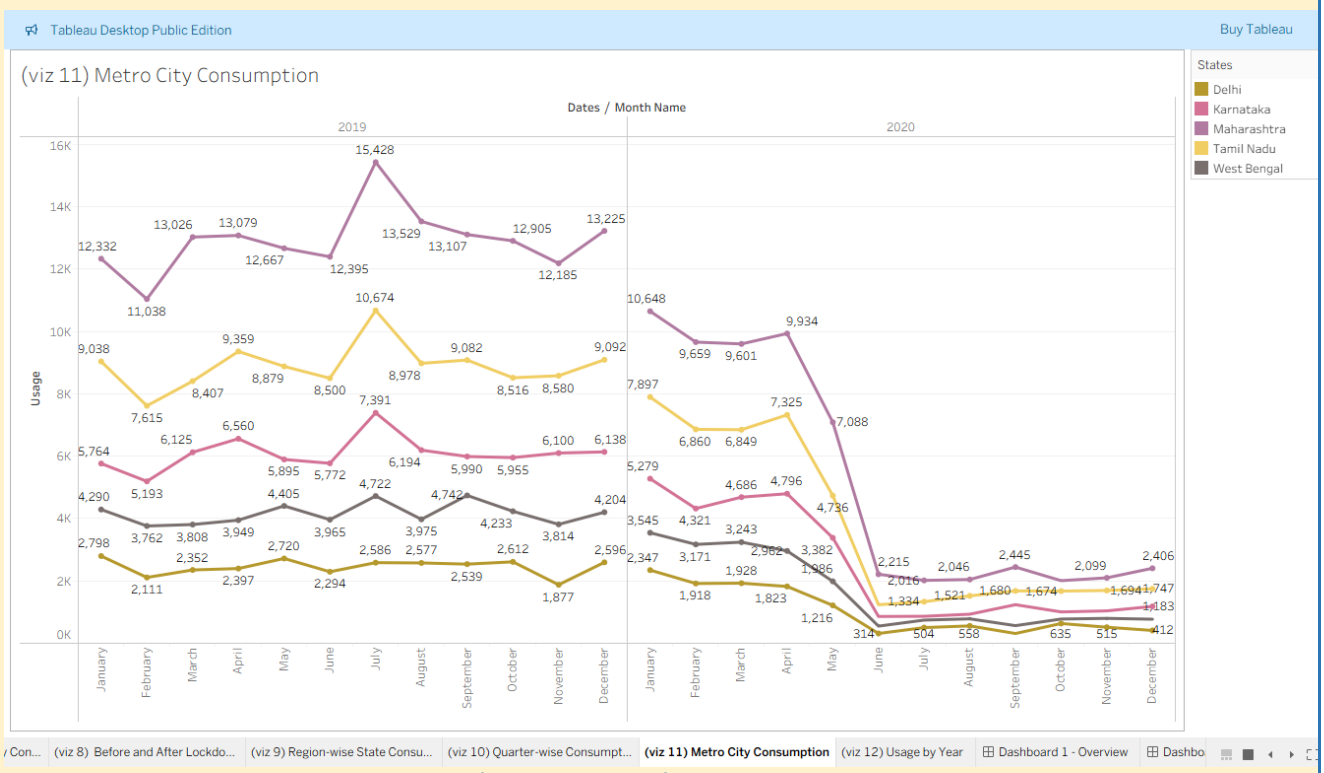
Field	Details
Chart Type	Bar Chart
X-axis	Quarter (Q1, Q2, Q3, Q4)
Y-axis	Total Usage (MU)
Colour	Year (2019 vs 2020)
Key Insight	Q3 2019 is the peak quarter (July–Sep = monsoon industrial demand). Q2 2020 shows the COVID collapse.
Assigned To	Ishita Prajapati



Viz 10 — Quarter-wise Usage

Visualization 11 — Metro City State Usage

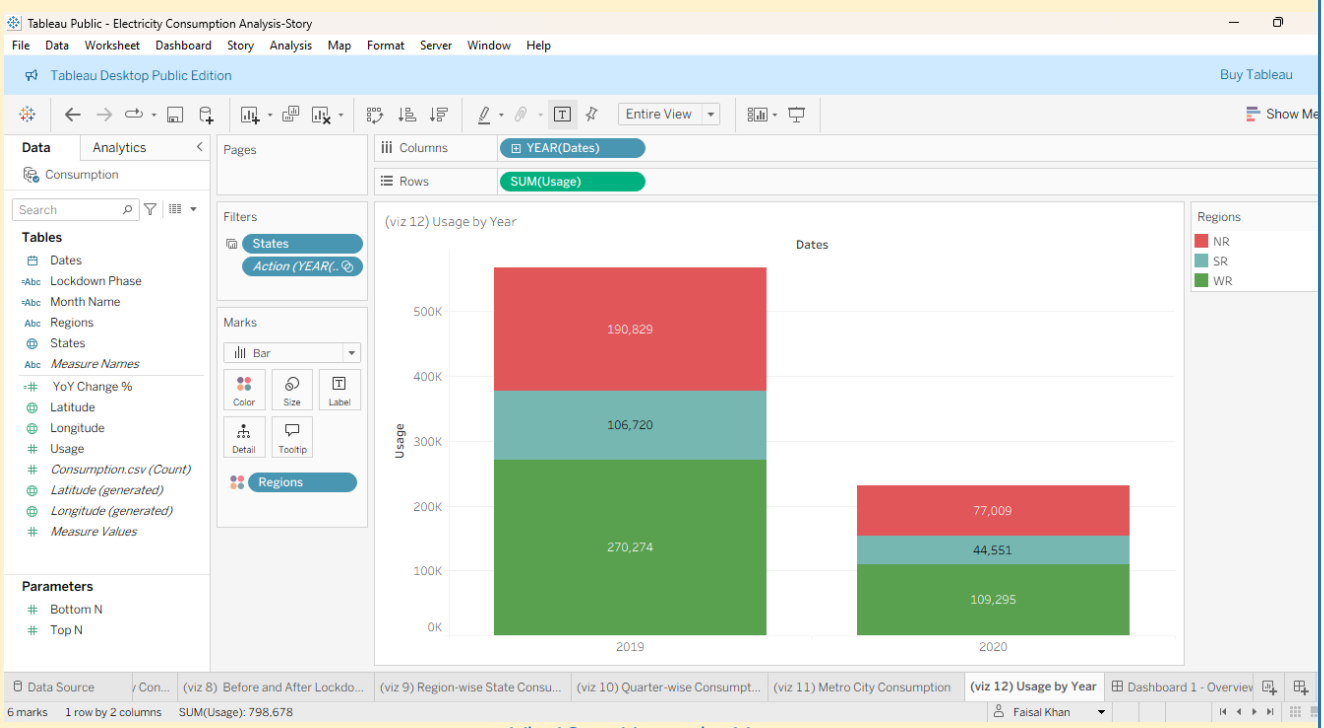
Field	Details
Chart Type	Bar Chart
States	Delhi, Maharashtra, Karnataka, Tamil Nadu, West Bengal (major metro states)
Measure	SUM(Usage) by month
Key Insight	Delhi showed sharper percentage drop during lockdown compared to Maharashtra due to its smaller industrial base.
Assigned To	Faisal Khan



Viz 11 — Metro City State Usage

Visualization 12 — Usage by Year

Field	Details
Chart Type	Side-by-side Bar Chart
X-axis	Year (2019 vs 2020)
Y-axis	Total Usage (MU)
Filter	State or Region (optional)
Key Insight	2019 total: 1,216,895 MU. 2020 total (partial): 417,521 MU. On a monthly average basis, 2020 is ~65% of 2019.
Assigned To	Kali Bakshi



Viz 12 — Usage by Year